INCH-POUND
MIL-DTL-24643/68B
27 November 2023
SUPERSEDING
MIL-DTL-24643/68A
w/INT. AMENDMENT 1
11 July 2011
MIL-DTL-24643/68A
28 January 2010

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +221 DEGREES FAHRENHEIT (+105 DEGREES CELSIUS), 1,000 VOLTS, TYPE LS4OW

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction (Watertight with Circuit Integrity)

First	-	Copper conductor, coated or uncoated ASTM B8 (see table I for size).
Second	-	Thermoset insulation, extruded or taped (see table I for wall thickness).
Third	-	An optional glass braid.
Fourth	-	An optional covering, unless required (see 3.3.6 of MIL-DTL-24643).
Fifth	-	Four conductors shall be cabled together with a lay in accordance with 3.4.4 of MIL-DTL-24643. Cabling sequence shall be consecutive. Fillers may be used as necessary to form a firm, well-rounded assembly. Standard identification code applied by method 1 or 3, or letter identification code applied by method 5. If identification method 1 is used, outer surface to be printed shall appear black.
Sixth	-	Aluminum foil/polyester tape/aluminum foil. 0.002-inch thick with 45 percent overlap.
Seventh	-	Braided shield of tin-coated copper.
Eighth	-	An optional binder.
Ninth	-	Cross-linked polyolefin jacket. See table I for wall thickness.

NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

Distribution Statement A. Approved for public release: distribution is unlimited.

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TABLE I. Details.

Military part no. M24643/68	Type and size	Conductor size (AWG)	Conductor resistance (ohms)	Insulation resistance (megohms)		Insulation wall thickness min. avg. (inch)	Jacket wall thickness min. avg. (inch)	Overall diameter (inches)		Accelerated service loading (amperes)
				Conductor to conductor	Conductor to shield			min.	max.	
-01UO	LS4OW-3	16 (class B)	4.3	500	250	0.018	0.027	0.463	0.497	--
-02UO	LS4OW-4	14 (class B)	2.68	500	250	0.018	0.028	0.485	0.521	--
-03UO	LS4OW-9	10 (class B)	1.08	500	250	0.018	0.036	0.588	0.631	--
-04UO	LS4OW-14	9 (class B)	0.859	500	250	0.018	0.040	0.772	0.828	90
-05UO	LS4OW-23	7 (class B)	0.543	500	250	0.030	0.050	0.900	0.965	115
-06UO	LS4OW-50	3 (class C)	0.21	200	100	0.030	0.050	1.084	1.163	185
-07UO	LS4OW-75	1 (class C)	0.134	175	85	0.035	0.050	1.254	1.345	250
-08UO	LS4OW-100	0 (class D)	0.106	160	80	0.035	0.060	1.396	1.498	295
-085UO	LS4OW-150	000 (class D)	0.0674	135	65	0.040	0.060	1.600	1.725	400
-09UO	LS4OW-200	0000 (class D)	0.053	125	65	0.050	0.060	1.850	1.984	465

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REQUIREMENTS:

Qualification required.

INSPECTION:

Basic electricals:

Conductor resistance (ohms/1,000 feet at 77 °F [25 °C], max.)	See table I
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor (sizes 3 through 9)	3,000
Conductor to conductor (sizes 14 through 200)	5,000
Conductor to overall shield (sizes 3 through 9)	1,500
Conductor to overall shield (sizes 14 through 200)	2,500
Insulation resistance (megohms/1,000 feet, min.)	
Conductor to conductor	See table I
Conductor to overall shield	See table I
Conductor and shield continuity	No failure
Jacket flaws	No failure

Group A:

Visual and dimensional	No failure
Watertightness (see MIL-DTL-24643 for limits of water leakage)	No failure
Crack resistance (applicable to constructions using glass braids with coverings)	No damage
Abrasion resistance – scrapes minimum (size 23 and smaller) (extruded insulation only)	250

Group B:

Thermoset proof test (percent, max.)	
Insulation (extruded insulation only)	50
Jacket (when tested at 392 °F [200 °C])	50
Drip (203±1.8 °F [95±1 °C]) (see MIL-DTL-24643 for limits of drip allowance)	No failure
Cold bending (cable)	No failure
Physicals (unaged)	
Insulation (extruded insulation only)	
Tensile strength (lb/in^2, min.)	700

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Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1,300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35

Group C:

Physicals (aged) (air oven)	
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Shrinkage	No failure
Heat distortion (percent of unaged, max.)	30
Permanence of printing (insulation – method 1 only) (cycles, min.)	25
Permanence of printing (jacket) (cycles, min.)	125
Cable fill (sealant) removability	No failure
Surface transfer impedance	
Milliohms per meter (max.)	700
Electromagnetic pulse (EMP) response (dB, min.)	60

Group D:

Gas flame (circuit integrity) (1 hour)	No failure
Flame propagation (cable)	No failure

Qualification inspection:

Qualification inspection shall include basic electricals; Groups A, B, C, and D; plus the following:

Accelerated service (sizes 14 and larger) (see table I for load current)	No failure
Cold working (-4±3.6 °F [-20±2 °C])	Not applicable
Gas flame (circuit integrity) (3 hours)	No failure
Aging and compatibility (cable) (257±9 °F [125±5 °C])	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2

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Fillers	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) – type A (hardness, min.)	80
Weathering (jacket)	No failure
Electrical moisture absorption (extruded insulation only)	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LS4OW (all sizes)	500

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2013-084)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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